



UNIVERSITY OF CENTRAL PUNJAB LAHORE

Assignment # 1 Version 1 CLO Mapping: CLO1

Course Code	MAT243	Semester	Fall 2025
Course Title	Multivariable Calculus		
Resource Person	Dr. Muhammad Rayees Ahmad		
Assignment Given Date	25-10-2025	Submission Date	28-10-2025
Total Marks	25= Assignment (20)+Quiz based Viva (5)		

Version	Name of student	Complete Registration no.	Section	Assignment marks	Viva marks	Total marks
<u>1</u>						
<u>2</u>						

Submission Instructions (Please follow strictly)

- Take the PRINTOUT OF THIS ASSIGNMENT PDF file and solve the question at the given empty spaces in the assignment PDF file. DO NOT use any other papers. Alternatively, you can only use simple plane white printer pages.

- This is an **INDIVIDUAL ASSIGNMENT**.
- Version 1 will solve the students who has the last digit ODD in their registration number.
- Version 2 will solve the students who has the last digit EVEN in their registration number.
- It is mandatory to know the whole method of all versions. Any question may be asked in the viva.

- Write your complete names and registration numbers on each page of the assignment.
- Follow the deadline. Finish your work one day before, so you could submit in time.

- LATE SUBMISSION WILL NOT BE ACCEPTED (Deduction of 5% marks each day).
- All Plagiarized assignments will be awarded ZERO marks.

I declare that I have prepared the assignment according to above guidelines, and I shall be responsible for any deduction of marks if the instructions are not followed.

Student Signature: _____

Version: 1**Question [CLO1] (20 marks=8+5+5+2):**

A **robot arm** in a manufacturing cell moves along the **line** passing through the points $A(2, -1, 3)$ and $B(6, 2, 7)$. A sensor is located at the point $P(4, 3, 5)$.

- a) **Find** the **vector and parametric equations** of the robot arm's line of motion.
- b) **Plot** the equation on the **GeoGebra Software** and attach the print.
- c) **Find** the **perpendicular distance** from the sensor P to this line using

$$d = \frac{|\vec{u} \times \vec{v}|}{|\vec{v}|}$$

- d) Comment on whether the sensor is within 3 units of the arm's path.

Marks Distribution: part (a): 8 marks, part (b): 5 marks, part (c): 5 marks, part (d): 2 marks.

Solution:

Name:

Registration No.

Name:

Registration No.



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Assignment # 1 Version 2 CLO Mapping: CLO1

Course Code	MAT243	Semester	Fall 2025
Course Title	Multivariable Calculus		
Resource Person	Dr. Muhammad Rayees Ahmad		
Assignment Given Date	25-10-2025	Submission Date	28-10-2025
Total Marks	25= Assignment (20)+Quiz based Viva (5)		

Version	Name of student	Complete Registration no.	Section	Assignment marks	Viva marks	Total marks
<u>1</u>						
<u>2</u>						

Submission Instructions (Please follow strictly)

- Take the PRINTOUT OF THIS ASSIGNMENT PDF file and solve the question at the given empty spaces in the assignment PDF file. DO NOT use any other papers. Alternatively, you can only use simple plane white printer pages.

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- Version 2 will solve the students who has the last digit EVEN in their registration number.
- It is mandatory to know the whole method of all versions. Any question may be asked in the viva.

- Write your complete names and registration numbers on each page of the assignment.
- Follow the deadline. Finish your work one day before, so you could submit in time.

- LATE SUBMISSION WILL NOT BE ACCEPTED (Deduction of 5% marks each day).
- All Plagiarized assignments will be awarded ZERO marks.

I declare that I have prepared the assignment according to above guidelines, and I shall be responsible for any deduction of marks if the instructions are not followed.

Student Signature: _____

Version: 2**Question [CLO1] (20 marks=7+5+6+2):**

A **data-collection drone** travels near an atmospheric plane. The **plane** passes through the **point** $Q(1, 2, 3)$ and is perpendicular to the vector $\vec{n} = (3, -2, 6)$. A drone is at position $R(4, 5, 9)$.

- Find** the **equation of the plane**.
- Draw** the equation of plane on **GeoGebra Software** and attach the print
- Find** the perpendicular distance from R to the plane using the formula

$$d = \frac{|\vec{u} \cdot \vec{n}|}{|\vec{n}|}$$

where $\vec{u} = \overrightarrow{QR}$.

- Check if the drone lies within 4 units of the plane.

Marks Distribution: part (a): 7 marks, part (b): 5 marks, part (c): 6 marks, part (d): 2 marks.

Solution:

Name:

Registration No.

Name:

Registration No.
